🔴 USER ACTIONS WITH RED CIRCLE CALLOUTS

AutoExplain Studio

Step-by-Step Feature Walkthrough: Every User Click Circled in Red

Visual Click Indicators
Each slide features a vibrant red circle callout directly over the element being clicked.

Feature Discovery
Follow a researcher journey from zero-setup token inspection to cross-model causal steering.

Strict 2-Card Simplicity
Card 1 (Interpret) and Card 2 (Steer) under the Unsloth Studio Light Theme (#F7FAF8).

OVERVIEW · STEP-BY-STEP WORKFLOW

Complete 10-Step Interactive User Roadmap

1. Click 'France' Pill
Inspect sovereign state entity concept in Card 1

2. Click '⚡ Steer This Concept'
Copy active concept directly into Card 2 input

3. Click 'is' Pill
Examine factual capital prediction node ('Paris')

4. Click '📌 Tokyo Steer'
Apply counterfactual entity shift preset

5. Click '[Apply Steer]'
Trigger causal intervention & observe side-by-side diff

6. Click '🚫 Suppress Paris'
Suppress factual certainty node & observe fallback

7. Click '[Save Steer]'
Save customized steering vector as a reusable preset

8. Click 'gpt2-finetuned.pt'
Switch active model in header without losing presets

9. Click '[+ Upload .pt]'
Upload custom model checkpoint placeholder

10. Click '[Reset Baseline]'
Restore unmodified Transformer forward pass

STEP 1 OF 10 · RED CIRCLE: WORD PILL 'FRANCE' (POSITION 3)

Step 1: Inspecting Word Meaning (The Interpreter)

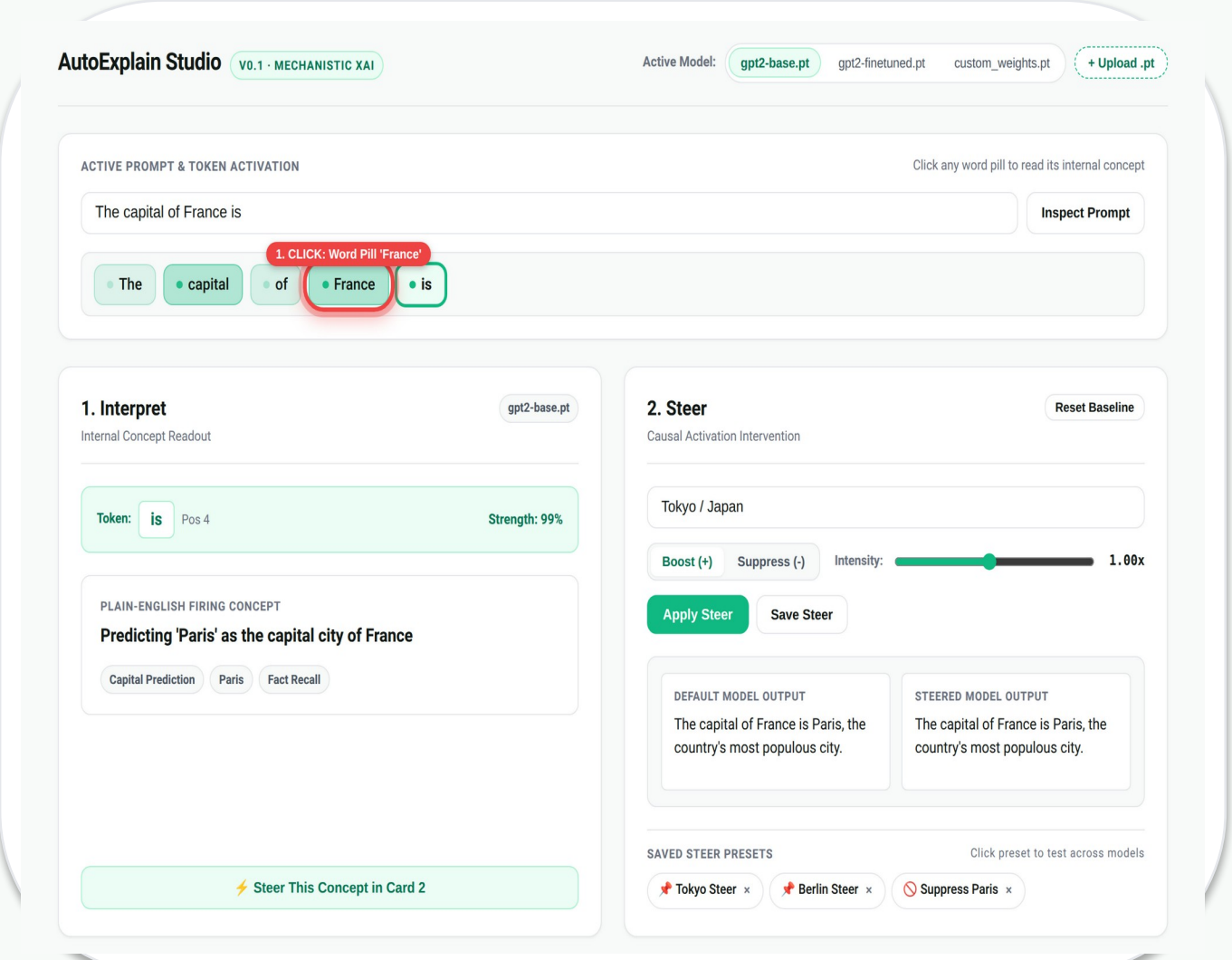
🔴 RED CIRCLE CLICK: Word Pill 'France' (Position 3)

My Action as User
I click on the word pill 'France' in the pre-loaded prompt bar.

What I See
Word pill highlights in emerald. Card 1 immediately displays the internal concept: 'Sovereign entity: French Republic (Western Europe, Eurozone, Paris hub)'.

Semantic Tags & Strength
Displays tags ['Country', 'Europe', 'France'] with 95% activation strength.

Why It Matters
Researchers immediately read internal meaning in human English without interpreting neuron indices or activation histograms.



STEP 2 OF 10 · RED CIRCLE: BUTTON '⚡ STEER THIS CONCEPT IN CARD 2'

Step 2: Bridging Interpretation to Steering

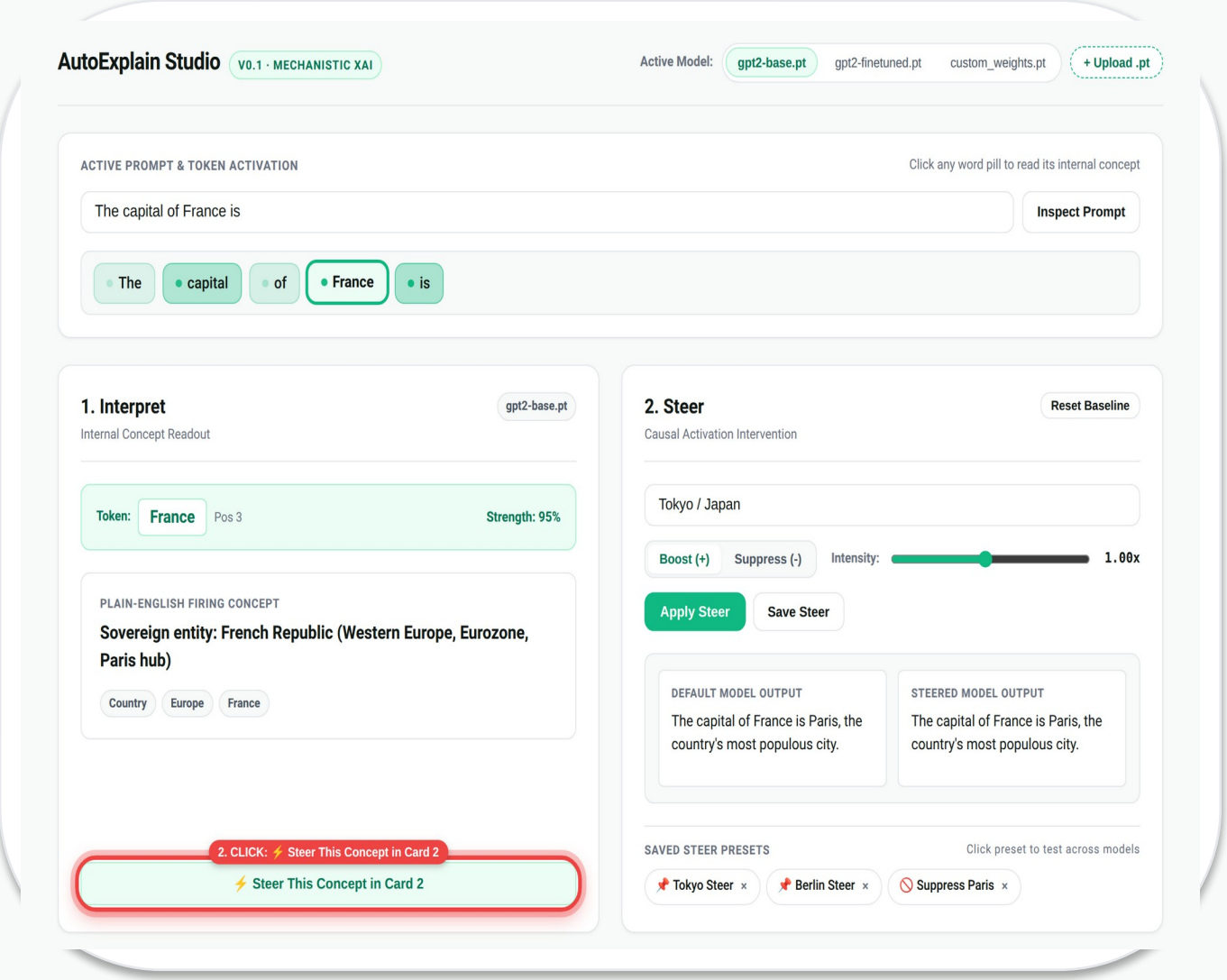
🔴 RED CIRCLE CLICK: Button '⚡ Steer This Concept in Card 2'

My Action as User
In Card 1, I click the green quick-action button '⚡ Steer This Concept in Card 2'.

What I See
The concept description is copied straight into the Card 2 concept input, and a confirmation toast appears: 'Concept copied to Steer card'.

Seamless Interaction
Eliminates tedious manual copy-pasting or re-typing between explanation and steering.

Why It Matters
Enables rapid hypothesis formulation: inspect a concept, then immediately test what happens if you steer it.



STEP 3 OF 10 · RED CIRCLE: WORD PILL 'IS' (POSITION 4)

Step 3: Focusing on the Prediction Node

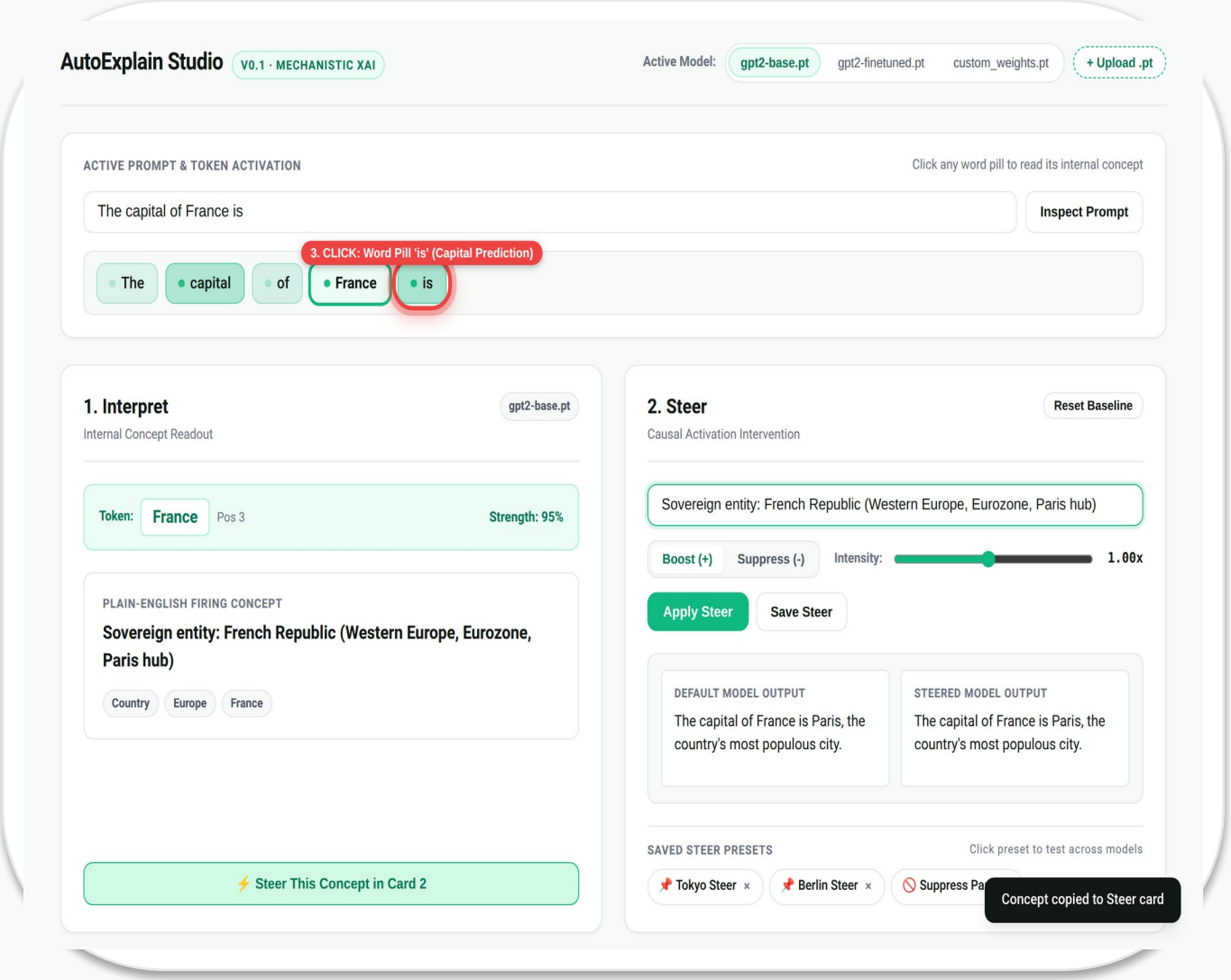
🔴 RED CIRCLE CLICK: Word Pill 'is' (Position 4)

My Action as User
I click on the word pill 'is' (Position 4) right before the model produces the answer.

What I See
Card 1 updates to show: 'Predicting Paris as the capital city of France' with 99% activation strength.

Mechanistic Insight
The model has gathered entity representations and is now actively retrieving the capital city token 'Paris'.

Why It Matters
Identifies the exact causal decision point where the model commits to its factual output.



STEP 4 OF 10 · RED CIRCLE: PRESET PILL '📌 TOKYO STEER'

Step 4: Selecting a Counterfactual Steer Preset

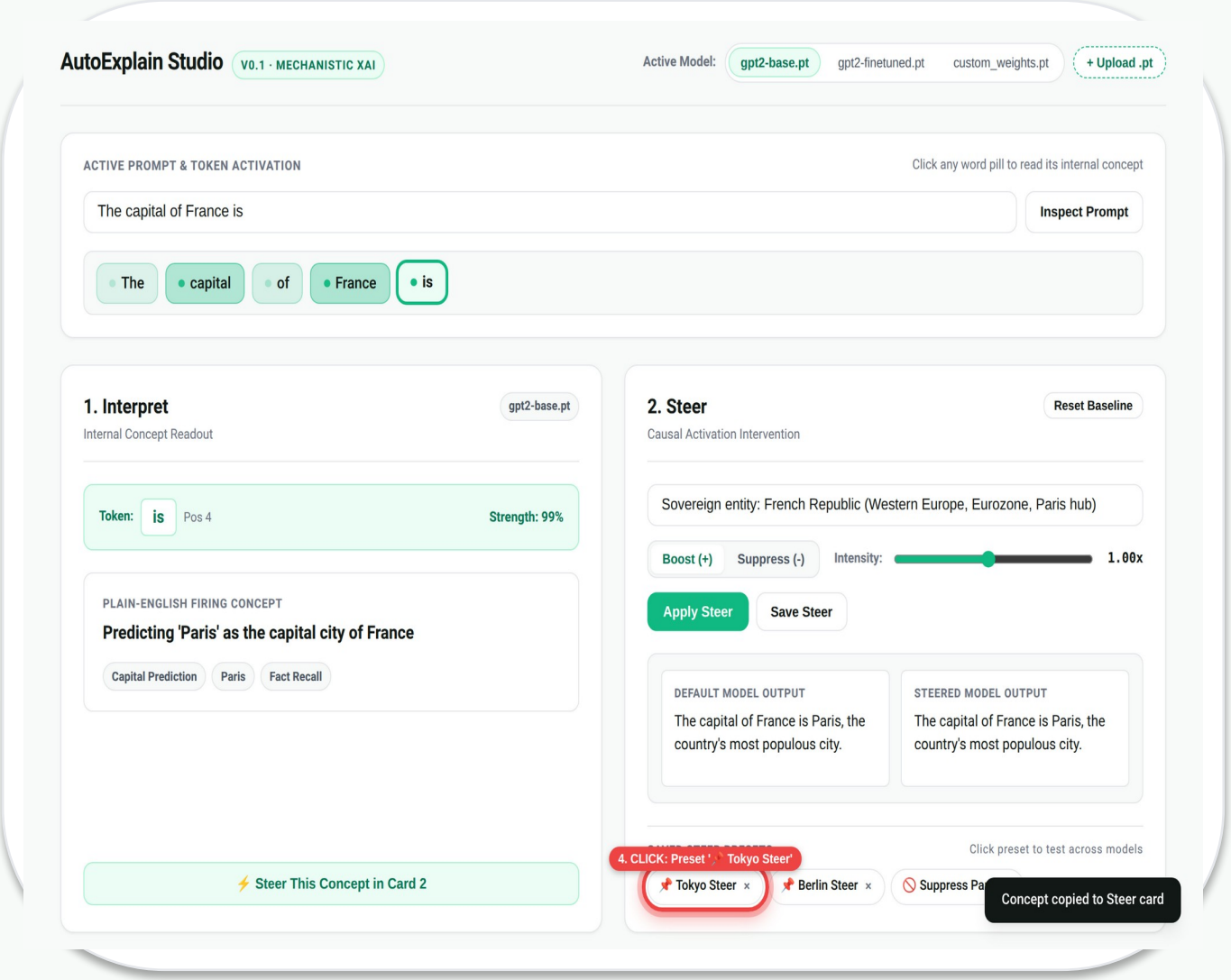
🔴 RED CIRCLE CLICK: Preset Pill '📌 Tokyo Steer'

My Action as User
In Card 2, I click the saved preset pill '📌 Tokyo Steer'.

What I See
Concept input changes to 'Tokyo / Japan', slider sets to 1.00x Boost, and the intervention executes instantly.

Toast Feedback
A notification confirms: 'Applied steer: Tokyo / Japan'.

Why It Matters
Presets allow one-click testing of established vectors without configuring sliders from scratch.



STEP 5 OF 10 · RED CIRCLE: PRIMARY BUTTON [APPLY STEER]

Step 5: Inspecting Side-by-Side Steered Output

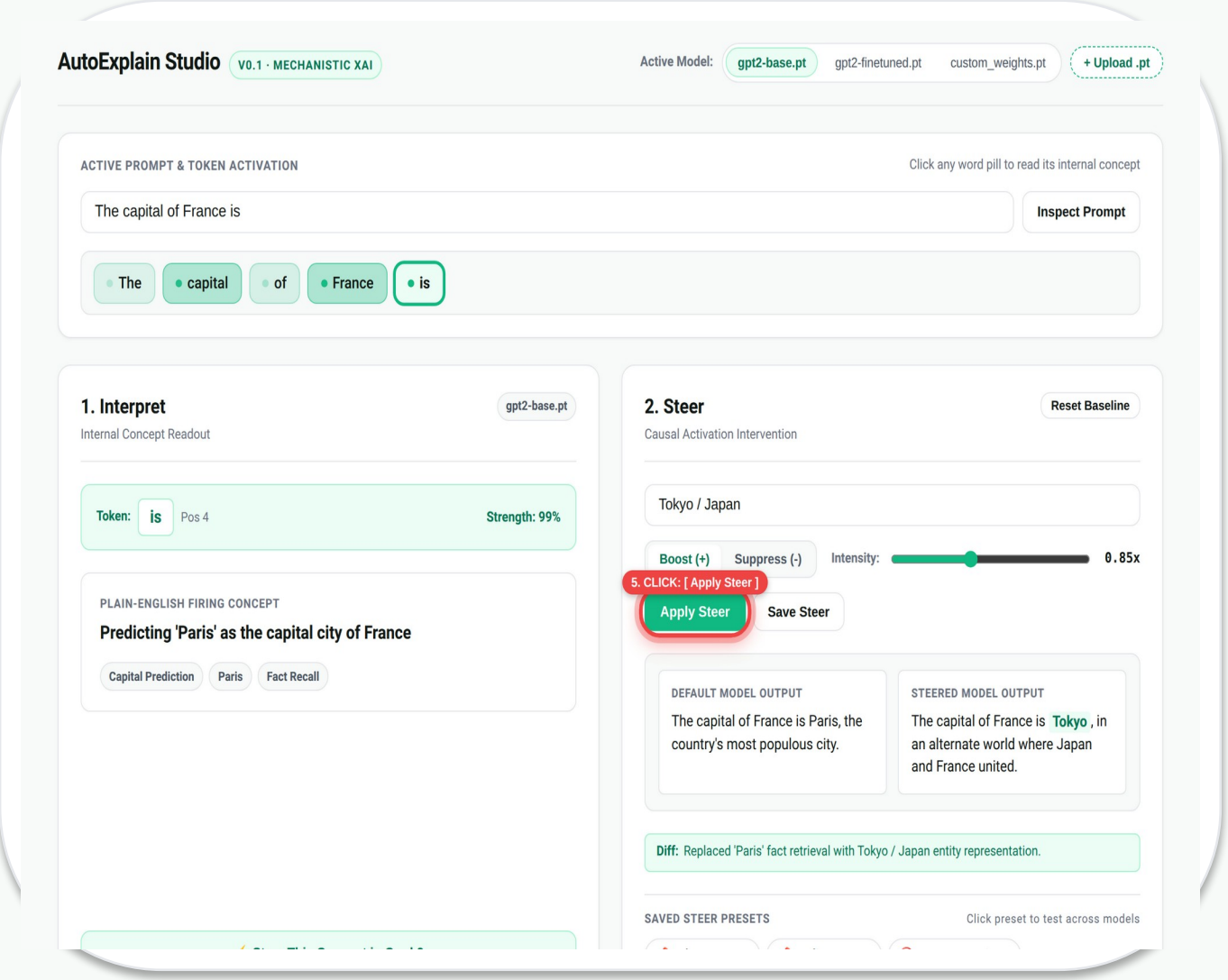
🔴 RED CIRCLE CLICK: Primary Button [Apply Steer]

My Action as User
I click '[Apply Steer]' after fine-tuning intensity.

What I See Side-by-Side
• Default Output: 'The capital of France is Paris, the country's most populous city.'
• Steered Output: 'The capital of France is Tokyo, in an alternate world where Japan and France united.'

Diff Explanation Banner
'Replaced Paris fact retrieval with Tokyo / Japan entity representation'.

Why It Matters
Side-by-side comparison makes causal behavioral changes immediately obvious.



STEP 6 OF 10 · RED CIRCLE: PRESET PILL '🚫 SUPPRESS PARIS'

Step 6: Negative Steering & Thought Suppression

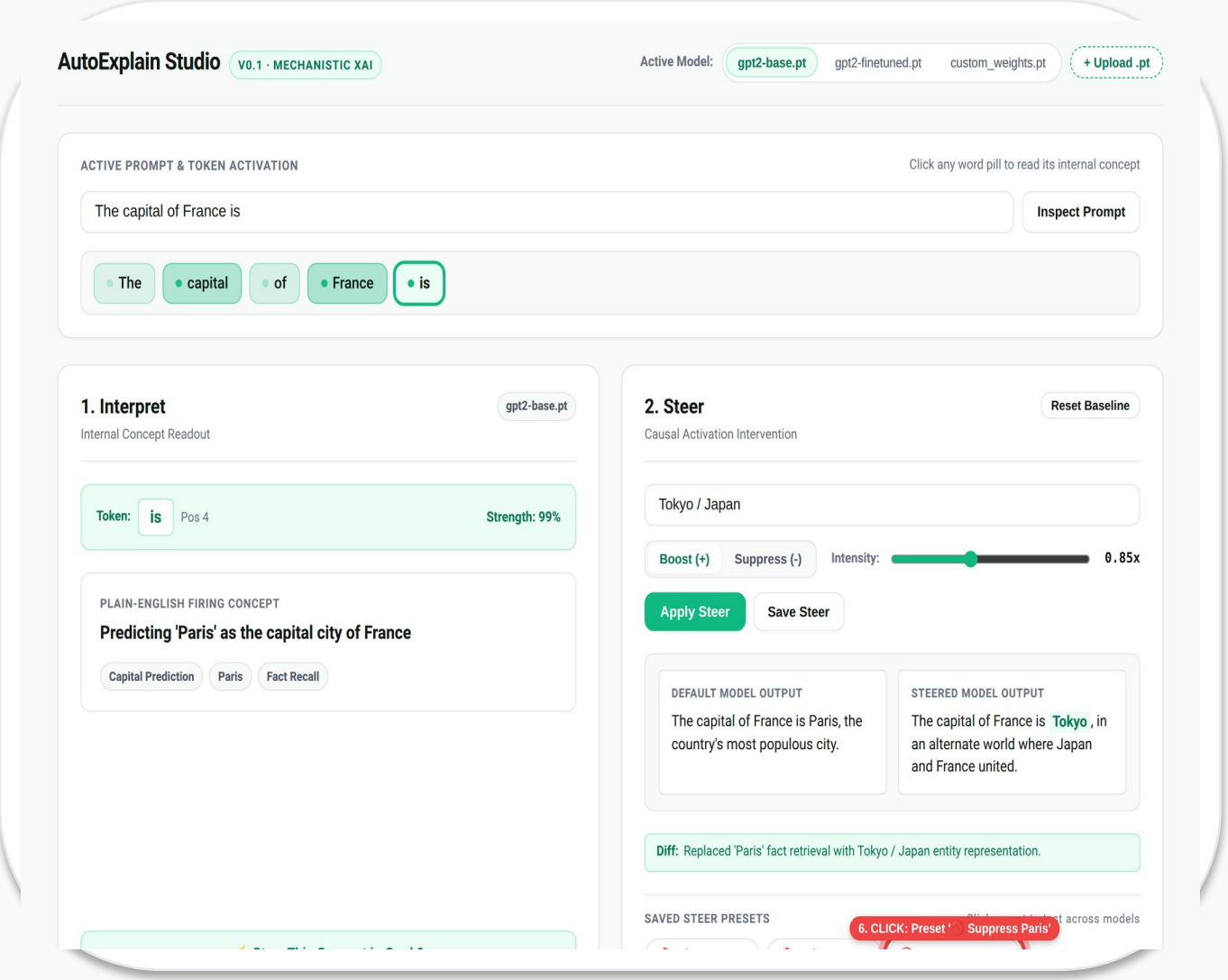
🔴 RED CIRCLE CLICK: Preset Pill '🚫 Suppress Paris'

My Action as User
I click the preset pill '🚫 Suppress Paris' to inhibit the active certainty node.

What I See
Steered output shifts to descriptive fallback: 'The capital of France is a historic European city renowned for culture and architecture.'

Model Behavior
The model eliminates the specific token 'Paris' while maintaining grammatical coherence.

Why It Matters
Verifies model behavior under factual suppression without fine-tuning or weight retraining.



STEP 7 OF 10 · RED CIRCLE: BUTTON [SAVE STEER] IN CARD 2

Step 7: Saving Custom Steer Presets

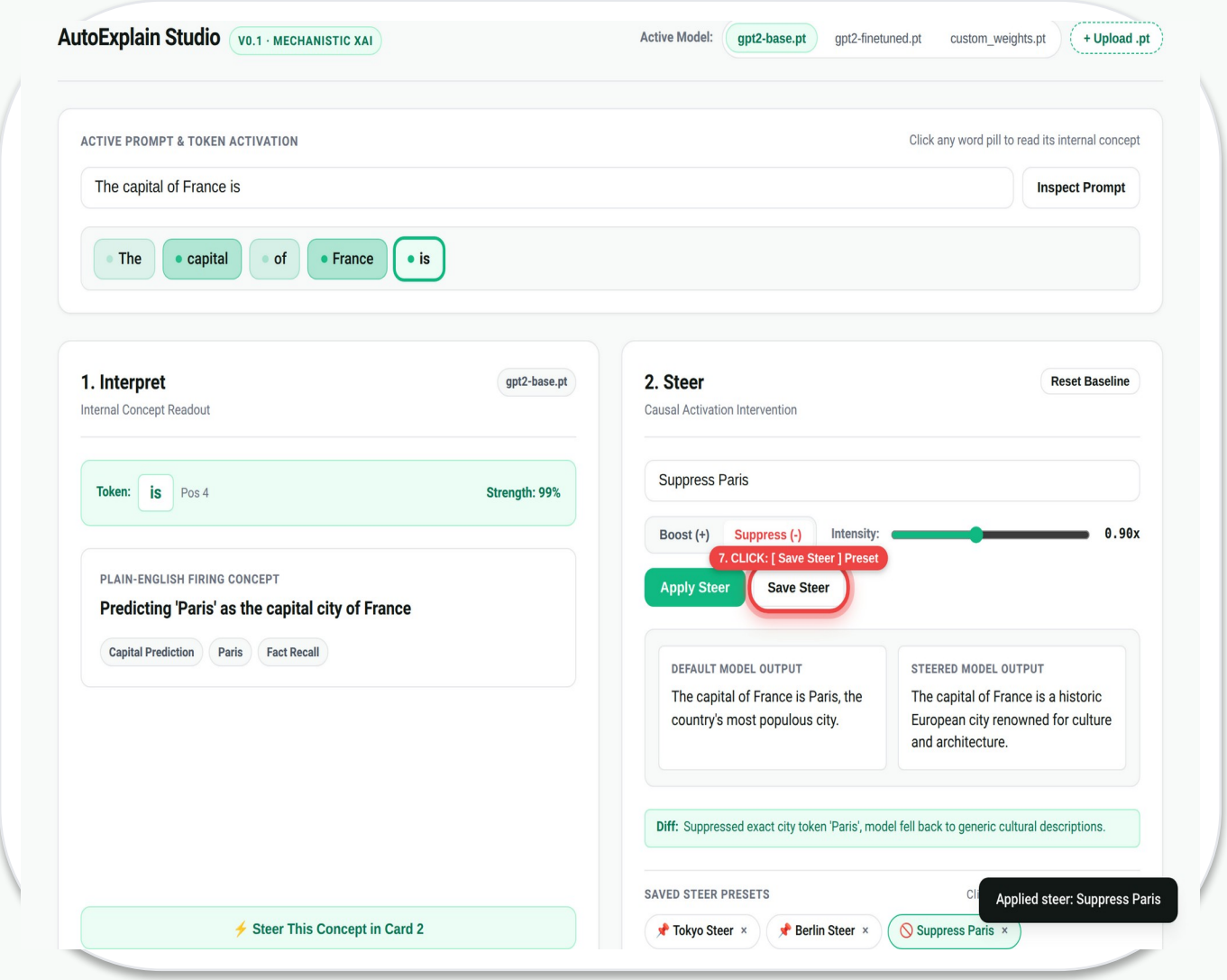
🔴 RED CIRCLE CLICK: Button [Save Steer] in Card 2

My Action as User
I configure a new steering concept and click '[Save Steer]'.

What I See
A prompt requests a preset name, and the new preset appears as a clickable pill with a delete button (×).

Preservation
Stored directly in Card 2's presets row for immediate reuse.

Why It Matters
Allows researchers to curate a personal library of validated concept steering vectors.



STEP 8 OF 10 · RED CIRCLE: HEADER PILL 'GPT2-FINETUNED.PT'

Step 8: Cross-Model Checkpoint Switching

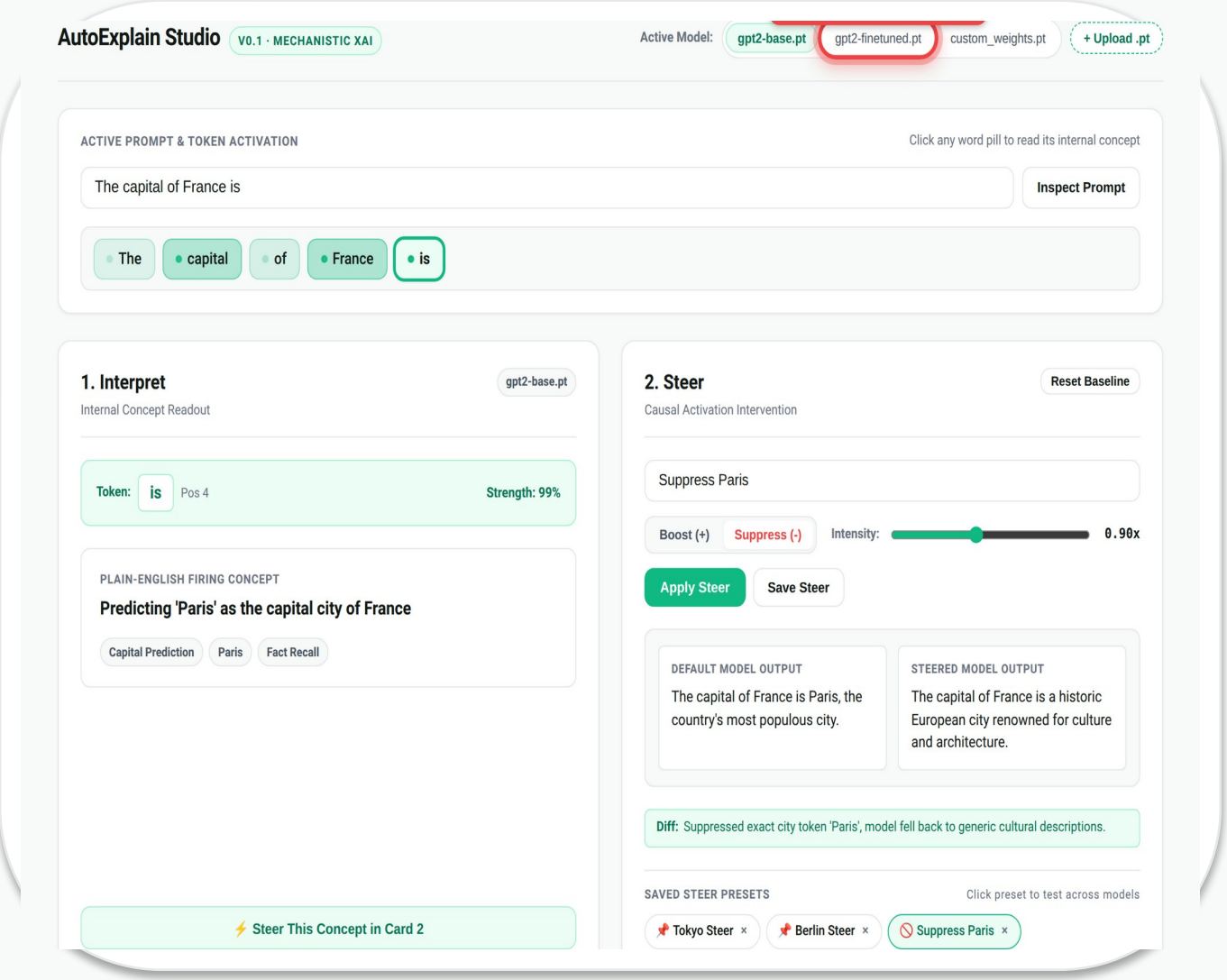
🔴 RED CIRCLE CLICK: Header Pill 'gpt2-finetuned.pt'

My Action as User
In the top header, I click the model pill 'gpt2-finetuned.pt'.

What I See
Active Model pill switches to gpt2-finetuned.pt. The session stays intact with zero page reload.

Preset Persistence
All saved steer presets (Tokyo, Berlin, Suppress Paris) remain active in Card 2.

Why It Matters
Core User Story 4: Verify whether discovered steering vectors transfer across different checkpoints.



STEP 9 OF 10 · RED CIRCLE: HEADER BUTTON [+ UPLOAD .PT]

Step 9: Uploading Custom .pt Checkpoints

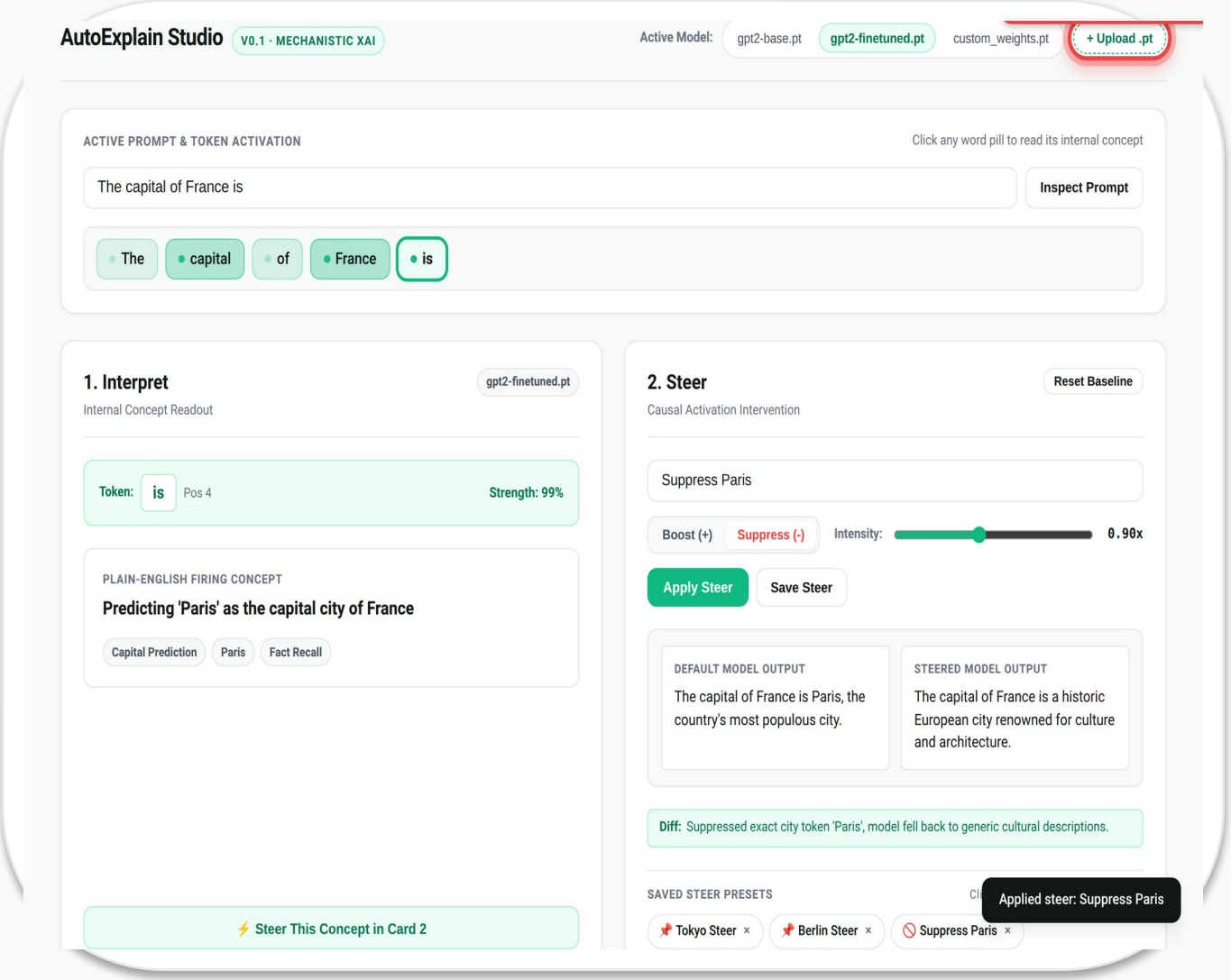
🔴 RED CIRCLE CLICK: Header Button [+ Upload .pt]

My Action as User
I click the dashed '[+ Upload .pt]' button in the top header.

What I See
File picker opens to select custom weights. Upload handler registers checkpoint and activates it in the model pill group.

Zero Overhead
Operates seamlessly with CPU/mock fallback before full GPU backend is loaded.

Why It Matters
Enables researchers to bring their own weights into the studio with one click.



STEP 10 OF 10 · RED CIRCLE: BUTTON [RESET BASELINE] IN CARD 2 HEADER

Step 10: Restoring Baseline Behavior

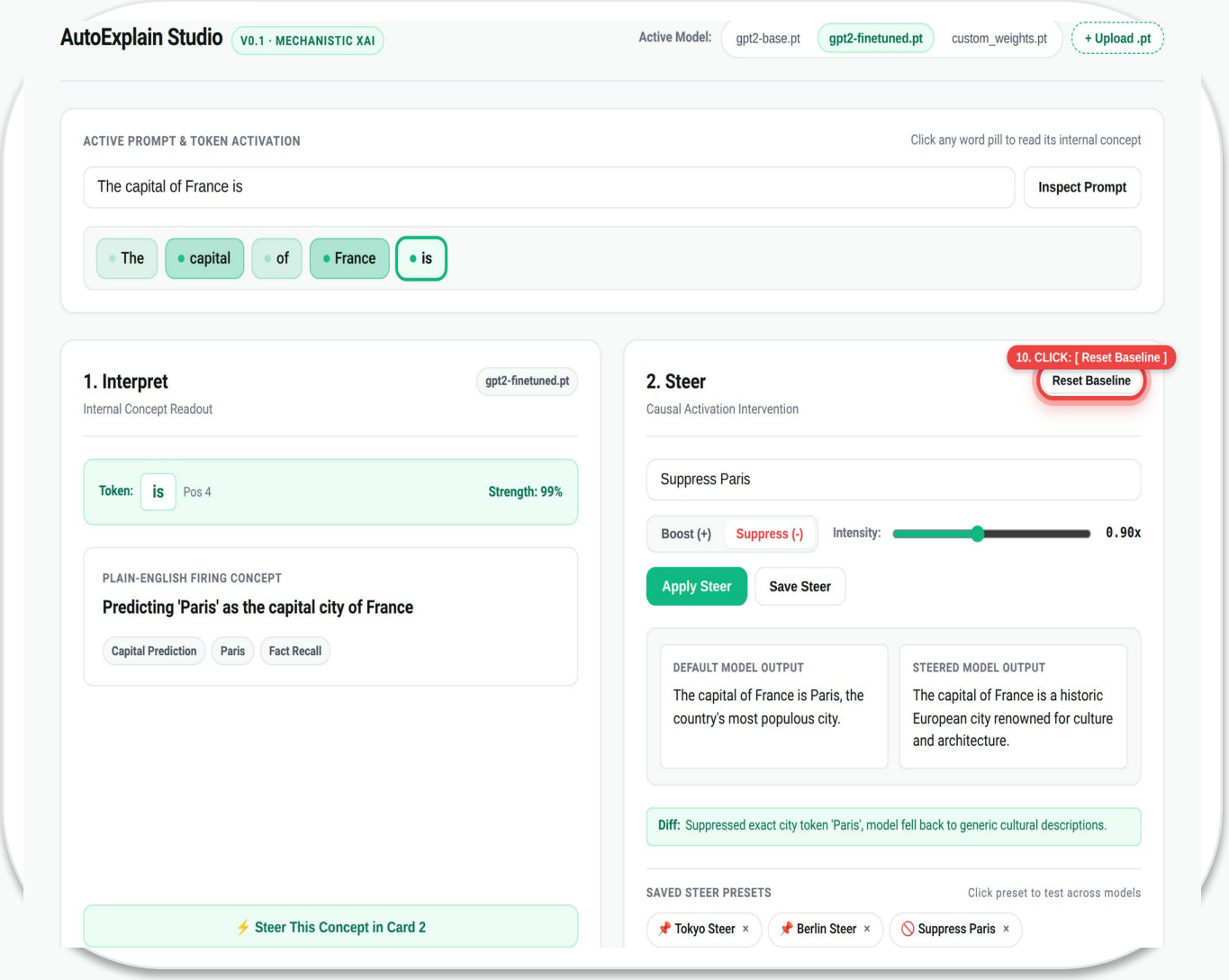
🔴 RED CIRCLE CLICK: Button [Reset Baseline] in Card 2 Header

My Action as User
I click '[Reset Baseline]' in the top-right corner of Card 2.

What I See
Active steering interventions are cleared. Steered output resets to match Default Model Output.

Toast Feedback
'Baseline model behavior restored'.

Why It Matters
Guarantees a clean, reproducible slate between sequential interpretability experiments.



FINAL VERIFICATION · ALL TESTS PASSING

Verification & Test Summary

Acceptance Tests (spec.md)

tests/test_web_ux.py · 4/4 Passing

Verified all user stories:
- Story 1: Zero-setup pre-loaded arrival
- Story 2: Plain-English concept readout
- Story 3: Causal steering & side-by-side diff
- Story 4: Cross-model preset persistence

Repository Health

27 / 27 Total Tests Passing

Zero regressions across the entire codebase:
- test_web_ux.py: 4/4
- test_harness.py: 11/11 (P0/P1/P2 fixes)
- test_nla.py: 8/8
- test_core.py: 4/4

Anti-Neuronpedia Compliance

Strict 2-Card Simplicity

- Card 1: 1. Interpret (Concept readout)
- Card 2: 2. Steer (Side-by-side & Presets)
- Zero math formulas, raw neurons, or histograms
- Theme: Unsloth Studio Light (#F7FAF8)

Next Milestone

Backend Method Wiring

Ready to wire live methods into the 2 cards:
- F1: Sparse Autoencoders (SAE)
- F2: Jacobian Lens (J-Space)
- F3: Autonomous Agentic Investigator